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On the expression of mistaken beliefs in Australian languages

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Abstract: This paper explores the expression of mistaken beliefs – as in for example *the boy mistakenly believes that the turtle is dead (whereas in fact it is alive)* – in the Indigenous languages of Australia. It is shown that some mode of expressing this meaning is attested in around 40% of the languages in a selection of 149 language varieties. In over 90% of the languages showing some mode of expressing the target meaning, it is – or can be – achieved through grammatical morphemes or constructions more or less dedicated to the expression of mistaken beliefs. These include particles, enclitics and various types of complement construction involving verbs of thinking, that frequently also convey meanings of saying, doing and hearing, rarely that specify the thought as mistaken. In just four or five languages, however, the meaning is attested only as a pragmatic implicature of a general statement of belief. To the extent possible given the limitations of the sources, the paper examines the range of meanings and uses of the morphemes/constructions expressing mistaken beliefs.

Keywords: Australian languages; counterfactuals; mistaken beliefs; modal particles/clitics; semblatives/similatives; thought complements

1 Introduction

Even children as young as five (indeed, much younger according to some investigators) are capable of understanding that someone else can hold a belief that conflicts with reality as the child knows it (e.g. Tomasello 2018). It seems likely that people everywhere will have occasion to communicate about such mismatches. Correspondingly, it seems reasonable to hypothesise that all languages provide their speakers with some means of asserting that such a mismatch obtains between a belief and understood reality. However, grammarians have paid scant attention to ways of expressing this notion, that is, that someone's belief at some point in time conflicts with the speaker's knowledge or beliefs.

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The present paper addresses this lacuna by undertaking a broad-based typological investigation of the means available in Indigenous Australian languages of expressing the meaning that someone's belief is mistaken. In some languages this meaning can be coded in a special morpheme or construction type dedicated to the expression of mistaken beliefs. This is the case, for instance, in Gooniyandi (Bunuban): the particle *thaarri* in (1) specifically marks the thought ('I took it sneakingly') attributed to the third person 'he' as mistaken according to the speaker's knowledge, and hence the gloss MB. To the best of my knowledge the mistaken belief meaning is inherent in such examples. No conflicting interpretation is possible, and the mistaken belief sense is not defeasible. Moreover, consistent with this, the proposition asserted as mistakenly believed can never be in the future with respect to the time of utterance: it must belong to the realm of past or present time.

Gooniyandi

(1) *niyi* *miga-mi* ***thaarri*** *yiganyi* *ward-la*
that say-3SG.NOM.CL **MB** uncertain go-1SG.NOM.3SG.ACC.CL
'He thought mistakenly "I took it sneakingly"', or 'He mistakenly thought he'd taken it sneakingly.' (McGregor 1990: 421)

By contrast, example (2) is neutral with respect to the truth or falsity of the thought (that the two men had been killed in retribution for their killing someone else). Although it turns out to be true, this is not the only possible interpretation. Example (2) could have been followed by a statement to the contrary (e.g. that they had been speared), and the mistaken belief meaning would then arise from context or by pragmatic implicature. Example (1), however, could never be followed by *yiganyi ward-nga* (sneakingly go-3SG.NOM.3SG.ACC.CL) 'he took it sneakingly'.

Gooniyandi

(2) *briyandi-yah/* *miga-winmi* *yaabja/* *briyandi-ya*
retribution-LOC say-3PL.NOM.CL some retribution-LOC
yood-binbidi-yi/
put-3PL.ACC.3PL.NOM-DU
'The others thought that the two of them had been put down (killed by magical means) in retribution.' (from the author's narrative corpus)

The aim of this paper is to typologise these expression types in Australian languages, addressing three main questions: (a) What means are available in the languages for making the mistaken belief meaning: what is the range of expression types? (b) What range of construction types are dedicated to coding mistaken beliefs, i.e. which expression types are grammatical signs coding this meaning? And (c) how are mistaken belief expressions used by speakers, and what meanings can they express

in discourse? The paper also touches briefly on theoretical questions concerning the analysis of the expression types and the relation between their forms and their meanings.

The main empirical claims of the paper are as follows. First, a number of Australian languages have a distinct construction type or types dedicated to the expression of mistaken beliefs. This can involve a morpheme such as a particle or enclitic that modalises the clause, coding its status as a mistaken belief. Such a morpheme can sometimes also occur in a complement construction. Second, some languages exclusively use a complement construction. This may be a distinct mistaken belief complement, or a general thought construction type that in certain contexts admits interpretation – by pragmatic implicature – as a mistaken belief.

This paper builds on an earlier paper on thought complements in Australian languages, McGregor (2021). That paper discussed various complement constructions, including reported thought constructions (as in e.g. *he thinks the turtle is dead*), a particular type of desiderative complement in which the desired event is represented as though a thought (as in *he thinks/says "I will climb the tree"*), and mistaken belief complements. The present paper elaborates and extends on the third of these topics, though it is not restricted to complement constructions; it includes all other attested modes of expression as well. It is also based on a considerably larger set of languages, just over twice as many as the earlier study, which was based on 73 languages. Otherwise, little has been published on mistaken belief constructions – except in descriptions of particular languages – and I am aware of no previous publication on their typology. As far as I am aware, just two conference presentations deal with the typology of mistaken belief constructions: Spronck (2017) which focuses on Australian languages, and Spronck and Vuillermet (2019) on Australian and South American languages.

The exposition is organised as follows. I begin in Section 2 by describing the selection of languages for the study. Following this, in Section 3, which comprises the bulk of the paper, we overview the range of modes of expressing mistaken beliefs in the chosen languages. Section 4 explores the semantics and pragmatics of mistaken belief expressions, to the extent possible given the inadequacies of the data. The paper concludes in Section 5 with an overview, some remarks on the limitations of the present investigation and suggestions for future research.

2 The language sample

The present investigation is based on a sample of 149 Indigenous Australian language varieties, selected from some 250–400 languages depending on one's criteria for

languagehood (e.g. Dixon 2002: 4–7; McGregor 2004: 1).[1] Thus, the sample represents between 37 and 60% of the languages. It is largely a convenience sample, selected with a few considerations in mind.

First, the language is ideally described in a modern grammar. Generally, only "comprehensive" grammars provide much relevant information, and a number of languages for which only sketch grammars were available were excluded from the sample.

Second, I aimed as far as possible to provide a reasonable genetic and geographical coverage. Twenty-three genetic lineages were represented from the 28–30 generally recognised families. However, to provide reasonably comprehensive geographical coverage it was necessary to include languages for which only pre-1960 grammars exist, many of which are sketch grammars. Even so, the sample remains unbalanced in favour of languages spoken in the northern part of the continent. These are currently the most viable languages, and most comprehensive modern grammars describe northern languages. This is also the region of greatest genetic diversity.

The approximate locations of the chosen languages are shown in Figure 1 below, which also indicates whether expression of mistaken belief is attested in the language, probable (though not entirely certain from the available information) or not attested. A full listing of the languages, organised genetically, together with a list of the sources consulted, is provided in the online Appendix. The Appendix also provides basic information on the mode of expression of mistaken belief (MB) in each language in which it is attested.

A few remarks are in order concerning the distribution of attestations of expressions of mistaken belief. To begin with, very few languages from the eastern states attest mistaken belief expressions. Those that do are mostly spoken in the far north, in the region of Cape York Peninsula. There are also a few languages located in the western parts of the eastern states. Thus Ngiyambaa (Pama-Nyungan), spoken in the central west of present New South Wales, is the only language of this state in which an expression is definitely attested. The southern part of the continent also shows a dearth of languages with attested mistaken belief expressions. These are regions that have had the most significant contact with Europeans, and the languages spoken in these regions were either already in very precarious states by the time they were first documented, or were documented in the nineteenth or early twentieth centuries. In fact, only in languages spoken in regions remote from major European settlement are expressions of mistaken belief attested – and even among these languages there are many that lack attestations of mistaken belief expressions (see

1 See also https://anggarrgoon.wordpress.com/2011/12/23/how-many-languages-were-spoken-in-australia/.

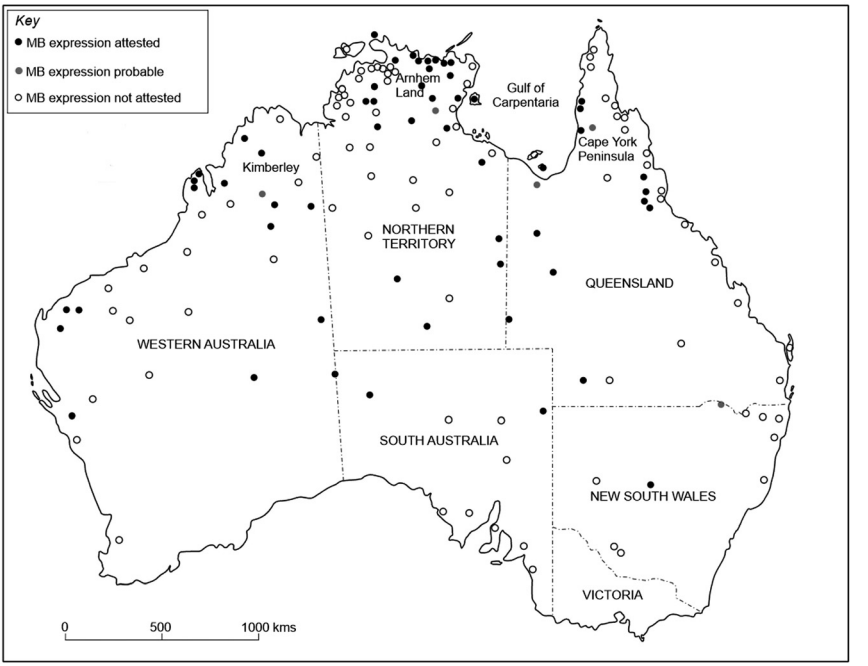


Figure 1: Map showing approximate locations of the languages in the sample and basic information concerning expressions of mistaken belief.

next section). These observations are not inconsistent with the hypothesised universality of the mistaken belief meaning, and absence of evidence of mistaken belief expressions in descriptions of languages should not be interpreted as evidence of their absence.

As indicated above, one of the main criteria for inclusion of a language in the database was the existence of a "comprehensive" modern grammar. Grammars were the main source of information for the present investigation. Other sources were also consulted when available, including dictionaries, wordlists and books and theses that focussed on aspects of the targeted language, especially if the comprehensive grammar was silent on mistaken beliefs. For example, for Murrinh-Patha (Southern Daly) I included Blythe's (2009) account of referring acts and Mujkic's (2013) description of the NP and clause since the grammar of Walsh (1976) provided no information. (Neither did the additional sources.) In most cases I did not search further into articles, the major exception being Breen (1984), an article specifically on mistaken belief constructions in Yandruwandha and Pitta-Pitta (both Pama-Nyungan).

3 Mistaken belief expressions: the range of possibilities

We have seen that a fair number of Australian languages have some means of expressing mistaken beliefs. They have some way of indicating the status of a proposition as the belief of some individual at some point of time – that in that person's opinion (at the time) the proposition was true – while at the same time maintaining that this proposition is in fact false (at least according to present speaker's knowledge). Such expressions provide a dual modal perspective on a proposition: they represent it as someone's belief whilst simultaneously asserting its falsity (*Represent* and *assert* are crucial in this formulation: see further Sections 4.4–4.6 below).

Often a single word represents the content presented as the mistaken belief, which concerns the categorisation of some entity at the centre of attention, as in the Mparntwe Arrernte (Pama-Nyungan) example (3). This mistaken categorisation sense is in fact the only one attested in some languages, e.g. Na-Kara (Maningrida) (Eather 1990: 343) and Ndjébbana (also Maningrida) (McKay 2000: 300). I would argue however that it is a full proposition that falls in the scope of the mistaken belief, a proposition concerning the categorisation of some entity, not just a nominal (or NP). To be precise, the nominal or NP that falls within the scope of the mistaken belief in such examples belongs to a verbless categorising clause in which the NP referring to the entity categorised has been omitted being given. Example (4) is one of the very few examples showing a non-elliptical categorising clause, *yarro jojarra* 'this is urine', comprising the definite NP *yarro* 'this' and a separate NP *jojarra* 'urine'.

Mparntwe Arrernte

(3) *arlenge-nge* *aherre-**kathene*** *ayenge* *itirre-ke*
far-ABL kangaroo-**MB** 1SG:NOM think-PC
arleye-rle
emu-FOC
'Hey! From afar I thought it was a kangaroo, but it turns out that it's an emu.'
(Wilkins 1989: 421)

Warrongo (Pama-Nyungan)

(4) *yarro* ***ngalnga-gaji*** *jojarra*
this **MB-MB** urine
'I mistakenly thought this was urine (but it was my semen)' (Tsunoda 2011: 677)

Mistaken belief expressions are attested in fifty-six languages; an additional six languages show expressions that can, with some degree of plausibility, be interpreted as

Table 1: Numbers of languages showing some mode of expression of mistaken beliefs.

Family	Number of languages with attestations of mistaken beliefs
Pama-Nyungan	33 of 86 languages + 3 uncertainties
Gunwinjguan	5 of 5 languages
Maningrida	4 of 4 languages
Nyulnyulan	4 of 6 languages
Arnhem	4 of 9 languages + 1 uncertainty
Worrorran	2 of 3 languages
Bunuban	1 of 2 languages + 1 uncertainty
Iwaidjan	1 of 2 languages
Tangkic	1 of 3 languages + 1 uncertainty
Isolate	1 of 1 language

conveying mistaken beliefs. These languages belong to ten genetic lineages, as shown in Table 1. In all, some mode of expressing a mistaken belief is attested in 38–42% of the language selection, and just over a third of the genetic lineages.

The most common way for uncertainties to arise is through imprecisions in translation or contextualisation. Consider the Bunuba (Bunuban) example (5). This seems a highly likely candidate for the mistaken belief interpretation, though this is not stipulated in the free translation. Given the free translation, it is not implausible that the speakers mistook the person for a tree. Furthermore, the mode of expression resembles a construction for expressing mistaken beliefs in the neighbouring, though genetically unrelated, languages Warrwa (Nyulnyulan) and Ngarinyin (Worrorran) – see Section 3.2.4 below.

Bunuba

(5) *lundu-yarra* *wara'ra* *yiyirrmiynhingi*
tree-PHP stand.up.3SG.NOM.CL.PRS 1EXC.NOM.say.PST.3SG.OBL
'We thought maybe he was a tree standing up there.' (Rumsey 2000: 104)

Sometimes the gloss provided for a morpheme suggests it could be used in expressing a mistaken belief, though this meaning is not mentioned or exemplified. Thus Ngalakgan (Arnhem) has particles *warwar* 'supposedly' and *yuw?we* 'supposedly' (Merlan 1983: 147), which one would expect to be employable in expressing a mistaken belief, though this sense is not mentioned. And indeed, the gloss 'supposedly' is used in some descriptions for a morpheme that clearly marks mistaken beliefs – including the descriptions of Burarra (Maningrida): Glasgow and Garner (1980) and Glasgow (1988).

Two major modes of encoding mistaken beliefs are attested in the languages of the sample: (1) a more or less dedicated morpheme, either bound or free; and (2) a

complement construction, usually in combination with a special marker. The mistaken belief meaning may also emerge as (3) a pragmatic implicature of a complement construction; occasionally this is the only attested mode of expressing the meaning. These three modes of expression are discussed in order in the following subsections.

3.1 Marked by a morpheme or collocation of morphemes

A fair number of languages possess a morpheme that either by itself or in a particular construction marks a proposition as a mistaken belief. This morpheme is usually a clitic or free particle (henceforth particle); rarely it is a suffix, and occasionally it is a collocation of morphemes. In the following subsections we discuss these possibilities in order, assuming the categorisation of the source; few grammars, however, mention or explicitly discuss the criteria for identification of the morpheme types. A language may employ more than one of these modes of marking mistaken beliefs; unfortunately, information does not permit postulation of meaning differences among the modes of expression.

3.1.1 A clitic marking 'mistakenly believe'

This is attested in 11–18 languages (about a fifth to a third of the languages attesting some mode of expression) belonging to three or four genetic families. The languages are widely scattered over the continent, as shown in Figure 2. Examples (6) and (7) are illustrative.

Enindhilyakwa (Gunwinjguan)

(6) *nuw-angkarree-ka* *erribaja-bu … wa,* *naree-ka* *amiyerra-**bu***
N-run.PST-EMP away-EMP … XTD NEG-EMP further-**MB**
'It ran further away, I thought it was going to keep going on, but no.' (van Egmond 2013: 373)

Kaytetye (Pama-Nyungan)

(7) *re* *areynengew-**athene*** *wenherre*
3SG euro-**MB** shot
'He thought it was a euro that he shot (but it wasn't).' (Turpin 2000: 114)

Examples in most of the languages reveal that the morpheme is a clitic that attaches to units of various types. Few descriptions, however, say anything about which unit actually hosts the enclitic. In the above examples it seems to be the focus of the mistaken belief. In (6) the distance travelled is the focus of the mistaken belief, and the clitic is attached to the adverbial *amiyerra* 'further'; in (7) it is the category of the



Figure 2: Location of languages with a possible enclitic marking mistaken belief.

thing that was shot that is at issue, not the shooting event itself which is evidently presumed to have occurred. However, a focus for the MB is sometimes lacking. For example, in (8) the MB does not focus contrastively on the speaker, but rather concerns the entire issue of their having gone for water. In this instance, the MB clitic appears in Wackernagel's position in the first clause.

Nyulnyul (Nyulnyulan)

(8) *ŋai-**elbe*** *ŋan-djed* *wol-oŋ,* *are*
*ngay-**ilbi*** *nga-ny-jid* *wul-ung* *arri*
1MIN.CRD-**MB** 1MIN.NOM-PST-go water-ALL not
ŋale-djedan
nga-li-jid-an
1MIN.NOM-IRR-go-IMP
'You mistakenly thought I went for water, but I did not.' (Nekes and Worms 1953: 519–520)

In some languages the clitic seems to be a dedicated marker of mistaken belief, at least in certain circumstances. This is the case, for instance, in Nyulnyul, Warrwa, Kaytetye, Mparntwe Arrernte, Yankunytjatjara (Pama-Nyungan) and Ngaanyatjarra (Pama-Nyungan), where the clitic seems to be used in no other sense and the mistaken belief sense appears not to be defeasible. Example (7) could not be followed by 'and it was', and nor could the final clause of (8) be replaced by the positive *nga-ny-jid* (1MIN.NOM-PST-go) 'I went'.

This is likely also the case in various other languages where it is accorded a different gloss, often modal in nature. For example, Kalkatungu (Pama-Nyungan) has what Blake (1979: 48–49) calls a prolocative suffix of uncertain meaning, though the examples he gives suggest it is an MB clitic. In Kugu Nganhcara (Pama-Nyungan) there is a clitic glossed 'supposedly', which suggests the MB interpretation, though no clear examples are given in Smith and Johnson (2000: 441). For Karajarri (Pama-Nyungan), Nekes and Worms (1953: 603, 2006: 292) identify a bound form that they refer to as a conjunction 'that'; examples given are consistent with its having MB as its only meaning, though it is attested only in complement constructions (see Section 3.2.1 below). And Warrongo shows what Tsunoda (2011: 682) refers to as a "counterfactual" clitic that usually has the MB sense; the exceptions warrant further investigation.

In four Pama-Nyungan languages – Arabana-Wangkangurru, Pitta-Pitta,[2] Yalarnnga and Yandruwandha – the clitic is either the semblative (i.e. similative, 'like a') marker (Breen 1984), or is homophonous with such a marker. (We return to this issue in Section 4.4 below.) In Pitta-Pitta the semblative clitic takes the form *-widi*, as in *yurathiyaka-widi* (shameless-SEM) 'as if (they had) no shame, shamelessly' (Breen 1984: 3). Examples (9) and (10) illustrate use of a clitic with the same phonological form in the expression of mistaken beliefs. (There are examples of *-widi* with other modal senses, though information is too scant for anything to be said with certainty.)

Pitta-Pitta

(9) *karnta-ya-**widi** inpa*
go-now-**MB** you
'I thought you were going. (But you aren't.)' (Breen 1984: 5)

Pitta-Pitta

(10) *kimpa-layi itji-ka-**widi***
alive-EMP die-did-**MB**
'He's still alive! We thought he'd died.' (Breen 1984: 6)

2 Blake and Breen (1973) consistently write the morpheme in Pitta-Pitta as a free word, though in Breen (1984) it is said to be a clitic.

The MB construction marked by the clitic does not specify the identity of the holder of the mistaken belief. In (6) and (9) it is the speaker, in (7) it is the third person, the shooter; in (8) the addressee; and in (10) a first person non-singular group. Context must be taken into account to determine (or guess) who the mistaken believer is. However, few descriptions discuss the issue, and for many languages it remains unclear whether or not there are genuine restrictions on who the mistaken believer can be. At least in some languages it is possible to specify the holder of the mistaken belief in a complement construction – see Section 3.2.1.

3.1.2 A (free) particle marking 'mistakenly believe'

This is the best attested expression type, showing up in 23–27 languages (around a half of the languages attesting an expression), as shown in the map in Figure 3.

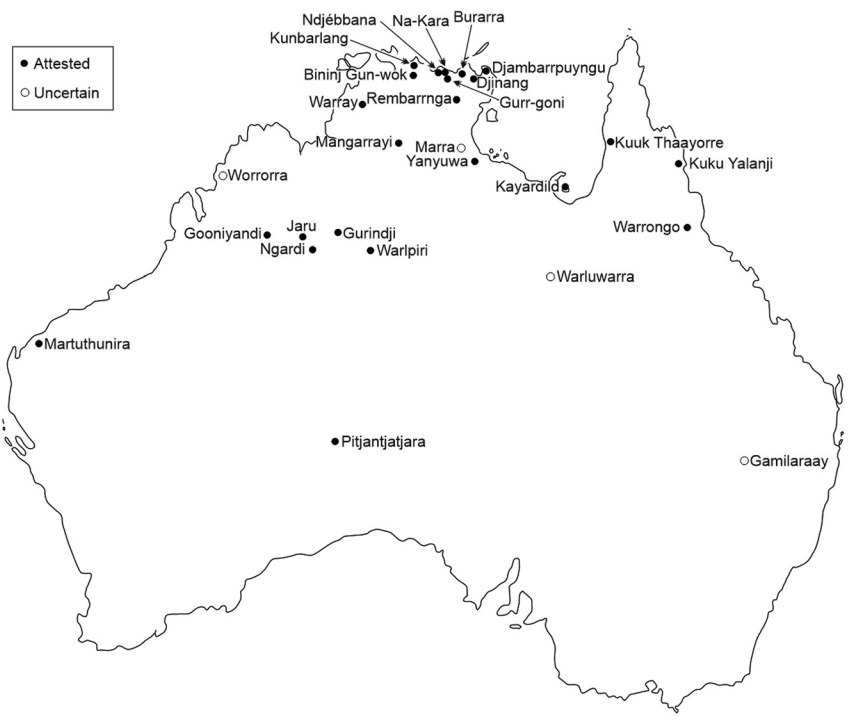


Figure 3: Location of languages showing a particle expressing mistaken belief.

Belonging to 6–7 families, the languages are again quite widely distributed geographically, though they tend to be more concentrated in the northern half of the continent.

Examples (11) and (12) illustrate this mode of expression.

Rembarrnga (Arnhem)

(11) ***guyala*** *yarra-many* *gurrah* ***guyala*** *jeny* *yarr-ma-ngœ-gan*
MB 1AUG-went that:way **MB** fish 1AUG>3-get-INF-DAT
'We went over there, we thought we were going to catch some fish (but we didn't).' (Saulwick 2003: 52)

Warrongo

(12) ***ngalnga*** *gogo* *goman*
MB language different
'I thought it was a different language (but in fact it was not).' (Tsunoda 2011: 677)

The position of the particle is rarely discussed in grammars, though as in the above examples it commonly occurs initially in the clause representing the mistaken belief.[3] This is the case also in Gooniyandi, where *thaarri* 'MB' normally precedes the unit it holds in its scope. However, the unit immediately following the particle can be the focus of the mistaken belief, as in (13), where there is contrastive focus is on the speaker and it is presupposed that someone hit someone.

Gooniyandi

(13) ***thaarri*** *nganyi-ngga* *gard-looni* *ngoorroo-ngga*
MB I-ERG hit-3SG.ACC.1SG.NOM.CL that-ERG
yaanya-ngga *gard-bini*
other-ERG hit-3SG.ACC.3SG.NOM.CL
'I thought I'd hit him, but really it was that other man who hit him.' (McGregor 1990: 425)

Rarely MB particles occur in some other position in the clause. In (14) it occurs between the initial NP (the Undergoer) and the verb; in (15) it occurs clause finally.

3 Example (11) is unusual in having another instance of the MB particle initially in the previous clause, which is not included in the scope of the MB particle. No explanation for this puzzling situation is provided in Saulwick (2003). Assuming it is not an error, it is reminiscent of what Haviland (1979: 154) has referred to as "ergative hopping", where an ergative marker sometimes occurs on the subject of an intransitive clause when it precedes a transitive clause.

Warray (Gunwinjguan)

(14) *wek* ***kunj*** *kat-tirim-pu-n* *pin* *amala*
firewood **MB** 1SG.NOM.POT-light-AUX-IRR but not
a-kutjkutj *amala* *kan-nu*
CL-wet not POT-burn
'I mistakenly thought I could light the firewood, but no, it was wet and wouldn't burn.' (Harvey 1986: 269)[4]

Kunwinjku (Gunwinjguan)

(15) *dja* *ngokko* *ø-bal-h-ka-ni* *na-wu* *mayh*
and already 3PST-away-IMM-carry-IMP MA-DEM animal
berrewoneng-ni ***yimankek****-ni*
them.two.OBL-IMP **MB**-IMP
'But already Echidna was bringing back food which she thought was for both of them.' (Evans 2003: 121)

Possibly in final position the particle serves as a type of afterthought. This interpretation is consistent with the second free translation given for the Warlpiri (Pama-Nyungan) example in (16).

Warlpiri

(16) *yuntardi* ***nganta***
beautiful **MB**
'She thinks she's beautiful (but).' Or 'She's beautiful, supposedly.' (Laughren 1982: 141; both free translations Laughren's)

Examination of the examples available reveals that the holder of the mistaken belief is most often the speaker (as in examples (12)–(14)). However, it can also be a first person group (example (11)) or someone else (examples (15) and (16)). I am unaware of any clear examples where it is the hearer, though this is almost certainly an accidental gap (examples are available for the clitic). Some descriptions state explicitly that context has to be taken into account to determine who the holder of the mistaken belief is, e.g. Evans (1995: 380) on Kayardild (Tangkic) and Laughren (1982: 149) on Warlpiri. Additionally, in some languages the holder of the mistaken belief may be specified by use of a complement construction – we return to this in Section 3.2.2.

In some languages the particle marking a belief as mistaken is dedicated to this use, which may be assumed coded by the particle. This seems to be the situation, for

4 Harvey gives a different free translation, but the one provided here is more consistent with what he says about the example in the ensuing text.

instance, in Gooniyandi, Mangarrayi (Arnhem), Jaru (Pama-Nyungan), Pitjantjatjara (Pama-Nyungan) and Djinang (Pama-Nyungan), where no other sense is attested for the relevant particle. Nor, to the best of my knowledge, is the mistaken sense defeasible in these languages.

Sometimes, however, the particle has other documented modal meanings and/or uses, typically in the domain of the counterfactual (which label is employed in some descriptions instead of mistaken belief). For instance, Gurr-Goni (Maningrida) has a modal particle *mundjarra* with two main senses in complementary distribution: (i) unrealised intention or expectation when the verb is in the irrealis; (ii) mistaken belief when the verb is in realis, or the clause is verbless (yielding the misclassification sense) (Green 1995: 337). Similarly, Na-Kara has a particle *ngamarramung* with meanings unfulfilled potential in the irrealis, and mistaken belief when preceding an NP (Eather 1990: 342); there is no mention of what happens with realis verbs. For Bininj Gun-Wok (Gunwinjguan) Evans (2003: 610–611) identifies a range of counterfactual meanings, which he says are associated with different constructions. Kayardild appears to be similar (Evans 1995: 378–379). In circumstances such as these, where the senses are in complementary distribution, there is presumably a single multifunctional modal particle and different constructions.

The different modal senses are not necessarily in complementary distribution in languages such as Ndjébbana (McKay 2000: 300), Kunbarlang (Gunwinjguan) (Kapitonov 2021; Kapitonov and Gentens 2018), and the Pama-Nyungan languages Marra (Heath 1981: 305), Kuku Yalanji (Hershberger and Hershberger 1982; Patz 2002: 113, 198, 220) and Martuthunira (Dench 1994: 168). It is not clear whether or not these are different senses of a single particle, or whether they are associated with different grammatical constructions.

In a few languages it seems that there may be, in addition to the MB particle, a homophonous particle with a modal meaning. Kuku Yalanji, according to Hershberger and Hershberger (1982), has two homophonous particles *balu*, one an MB particle (with various meanings, as per the previous paragraph), the other a negative particle 'not want'. In Tsunoda (2011: 677) identifies two distinct but homophonous particles *ngalnga*, one an MB particle, the other a prohibitive; interestingly, the MB particle almost always hosts the MB clitic, as in (4) above.

Like MB clitics, MB particles also encroach on the semblative domain, as discussed in more detail in Section 4.4 below. In a few languages, mistaken belief is marked by particles dubbed semblative markers. This is the case in Kuuk Thaayorre (Pama-Nyungan): examples (17) and (18) illustrate, respectively, the use of the free particle *kar* 'like' (as Gaby 2017 analyses it) in expressions of similarity and mistaken belief. Similarly, Kunbarlang has the particle *yimarne*, which Kapitonov and Gentens (2018) consistently gloss as a semblative, and which is also used in expressions of mistaken belief.

Kuuk Thaayorre

(17) *nhul* *koo.miing* ***kar*** *pam* *nhangnip* *nhangn*
3SG.NOM face **like** man father 3SG.GEN
'He looks like his father.' (literally, 'he is like his father in the face') (Gaby 2017: 328)

Kuuk Thaayorre

(18) *ngul* *nganip* *yik-r* *pul* *nganam-u* ***kar*** *parr_r*
then father say-P.PFV 3DU mother-ERG **MB** child
nhurr *thaa-munthi-rr*
2PL mouth-sink-P.PFV
'Then dad and mum said "(we thought) you kids had drowned."' (Gaby 2017: 449)

3.1.3 A suffix marking 'mistakenly believe'

There are a few languages with a mistaken belief suffix rather than an enclitic, though grammars do not always draw a clear distinction between the two. Ngaanyatjarra has in addition to an MB clitic also what Glass (1983: 40, 55) describes as a suffix that can occur on a nominalised form of a verb, on the 'relator' *palunya* 'that' and on a lexical nominal, as in (19). Indeed, the clitic *-lkanyu* MB may be simultaneously employed, attached to the initial word of the clause, as shown by (20), where it is hosted by the locative NP, while the suffix *-kukantja* MB attaches to a nominalised form of a verb.[5] It is not clear whether, and if so how, the suffix and enclitic contrast in meaning.

Ngaanyatjarra

(19) *wanti-ngu* *papa* *maru-**kukantja**-lu*
leave-PST dog black-**MB**-ERG
'He rejected it mistaking it for a black dog.' (Glass and Hackett 1970: 90)

Ngaanyatjarra

(20) *nyangka-rna* *ngurlu-ra* *wana-rnu* *pirti-ngka-**lkanyu***
and-1SG.NOM frighten-P.PTC follow-PST hole-LOC-**MB**
mapitja-ngu *tjarrpa-nytja-**kukantja**-lu*
go-PST enter-NOM-**MB**-ERG
'And I chased it mistakenly thinking that it had gone in and entered a hole.' (Glass 1983: 40)

5 Observe that the suffix precedes the case marker in (19), whereas the clitic follows the case marker in (20).

Another Western Desert variety, Pintupi (Pama-Nyungan), has an identical morpheme glossed 'mistakenly' in Hansen and Hansen (1978: 63), and is apparently also a suffix (it can be followed by case marking suffixes). The example provided is consistent with a mistaken belief interpretation. Marsh (1992: 101, 128) and Burgman (2009: 16, 21) also identify, but treat only cursorily, what might be mistaken belief suffixes in Martu Wangka (Pama-Nyungan), which could be cognate with the Ngaanyatjarra suffixes.

Another language that uses a suffix to mark mistaken beliefs is Ngiyambaa. This language employs the "circumstantive" suffix *-dji* CIRC, which otherwise marks certain types of ablative meanings such as prior existence, the standard of comparison (in comparatives)[6] and so on (Donaldson 1980: 96).

3.1.4 A collocation of morphemes

Mistaken beliefs are not always marked exclusively by a single morpheme. In some languages they are marked by a particular morpheme in a construction, which might be defined by the morpheme and a particular choice of verbal mood, as in Gurr-Goni and Na-Kara. Warrongo and Gurindji (Pama-Nyungan) have MB particles (see Section 3.1.2) that are normally accompanied by an enclitic: a counterfactual in Warrongo (according to Tsunoda 2011: 677) and a dubitive in Gurindji (Meakins and McConvell 2021: 550).

Two languages employ a combination of particles in the expression of mistaken belief. Wubuy ~ Nunggubuyu (Arnhem) uses the collocation *yingga muga*, where *yingga* is a particle meaning 'suddenly, imminent, nearly, almost' and *muga* a particle that Heath (1984: 438) says is "difficult to gloss, possibly some sort of mirative indicating surprise, sometimes doubt". Unfortunately, this pairing is very poorly represented in the source.

Kunjen ~ Oykangand (Pama-Nyungan) also uses two particles, the negative particle and another glossed 'suppose' (Sommer 1970: 239–240). In contrast with the situation in Wubuy, however, these occur discontinuously, apparently enveloping the focus of the mistaken belief, as shown by (21).

Kunjen ~ Oykangand

(21) ***ar*** *uw* *errmenham* ***oyong*** *il,* *abmanhdh* *amb*
not again finish.PST.PTC **suppose** she still PR
errgerrg *il*
speak.RED.PRS she
'I supposed she was finished, but she's still talking.' (Sommer 1970: 240)

6 As Jean-Christophe Verstraete (pers.comm.) has remarked, there is a possible link to the semblative sense remarked on above.

3.2 Realised by a complement construction

The second mode of expression of mistaken beliefs involves a complement construction. In some languages a marker (such as described in Section 3.1) may be optionally accompanied by a complement clause; in a few languages a complement clause is obligatory. McGregor (2021: 9–14) discusses the expression of mistaken beliefs in complement constructions; this section is an update.

The following types of matrix clause of thought are attested in mistaken belief complement constructions:

(a) with the vague (or polysemous) verb 'say, do, think';
(b) with the vague (or polysemous) verb 'hear, think';
(c) with a verb meaning 'think';
(d) with a verb meaning 'mistakenly think';
(e) with a nominal meaning 'mistaken belief'.

Mistaken beliefs are opinions about things, and as such are not (normally) addressed to anyone; the matrix clause never contains an NP specifying an addressee. Correspondingly, a second person category in the complement clause can only be interpreted as a second person in the utterance itself, as illustrated by example (22): the person who might have gone outside must be the addressee of the entire utterance. (See also examples (23) and (24).) The second person category is never according to the person origo of the mistaken thought.

Gooniyandi

(22) *miga-limi-nganggi* ***thaarri*** *banyangi* *ward-ginggi*
think-1SG.NOM.CL-2SG.OBL **MB** outside go-2SG.NOM.CL
'I mistakenly thought you'd gone outside.' (McGregor 1990: 421)

In many languages, however, the matrix clause can include an NP specifying the entity that the mistaken belief is about. This NP is often marked by the dative case, and in some languages is simultaneously cross-referenced by an oblique bound pronominal. The second person bound pronominal enclitic on the matrix verb in (22) cross-references such a participant, though the NP representing it has been ellipsed.

This section begins with a discussion of clitics, particles and suffixes in complement constructions (Sections 3.2.1–3.2.3); collocations of free particles are not attested in complement constructions. It then turns to a range of mistaken belief constructions obligatorily expressed by complement constructions (Sections 3.2.4–3.2.6).

3.2.1 A clitic with an optional matrix clause of thought

Almost all languages that attest an MB clitic admit the clitic in the complement of a matrix clause of thought. The only known exceptions are Warrongo and, if it has an MB clitic, Yukulta (Tangkic). On the other hand, the matrix clause of thought seems to be obligatory in Karajarri, given the examples presented in Nekes and Worms (1953, 2006).

The MB clitic is usually hosted by a unit in the complement clause, as shown by (23) and (24).

Nyulnyul

(23) *min-djed* *wol-oŋ* *djo-**elbe***
mi-ny-jid *wul-ung* *juy-**ilbi***
2MIN.NOM-PFV-go water-ALL 2MIN.CRD-**MB**
*ŋan-**d** dje*
*nga-n-**d**-jii*
1MIN.NOM-CM-**say**-2MIN.OBL
'I thought you went for water, but you did not.' (Nekes and Worms 1953: 519–520)

Yankunytjatjara

(24) *ngayulu* ***kuli**-nu* *nyuntu* *kutju-**palku*** *nyina-nyi*
1SG.NOM **think**-PST 2SG.NOM one-**MB** sit-PRS
'I thought, being under a misapprehension, that you were alone.' (Goddard 1985: 146)

This holds true also in the languages that employ a semblative clitic – or a homophonous morpheme – to mark the mistaken belief:

Pitta-Pitta

(25) *thartikanha-**widi*** *ngalilu* ***wangamaka***
water:hen-**MB** 1DU.EXC **know**.PST
'"We thought they were water hens."' (Given in answer to a question as to why they had killed the birds, which were in fact chickens.) (Breen 1984: 4)

Examples like these suggest that the clitic is hosted by the same unit it would have been hosted by in the absence of a complement clause (see Section 3.1.1).

In Warrwa, the MB clitic with a complement construction constitutes a distinct construction type from an ordinary thought complement. Briefly, the evidence is as follows (McGregor 2022a: Section 13.4.2.1). First, only the generic inflecting verb (IV) -JI 'say, do, think' can occur in the matrix clause; other verbs are permitted in ordinary

thought constructions. Second, the IV of thought invariably follows the mistakenly believed clause, as in (26) and (27); both orders are possible in ordinary thought constructions. Third, the IV of thought always appears on the same tone unit as the mistakenly believed clause, as shown by (26); in ordinary thought complements the two clauses often appear in separate tone units. Fourth, the matrix clause of thought is normally reduced, and typically comprises just the IV -JI 'say, do, think', with no overt NPs; if there is another unit in the matrix clause, it may be discontinuous with the IV, as in (27). In ordinary thought complements it is more common to find other NPs in the matrix clause, which is never discontinuous. Fifth, the complement clause invariably represents deictic categories of person, space and time in terms of the present speech situation; in no instance are they calculated from the origo of the actual thought event. For example, in (26) the past perfective and completive marking by *kaliya* 'finish, already' of the complement clause is in the past with respect to the thought event; they cannot be interpreted as past with respect to the time of the speech event and after the time of the thought. In the case of ordinary thought complements, both choices of origo are attested (with certain restrictions – see McGregor 2021: 6–7).

Warrwa

(26) *marlu* *kurd* *wi-la-rnda-na/* *kaliya-**mirrimba*** *kurd*
not dead 3NOM.IRR-IRR-go-PST finish-**MB** dead
ngi-rnda-ny *ø-**ji**-ny-jina/*
3MIN.NOM-go-PFV 3MIN.NOM-**say**-PFV-3MIN.OBL
'It's not dead. He mistakenly thought it was already dead.' (McGregor 2022a: 481)

Warrwa

(27) *ngayi-na* *yila-**mirrimba*** *nga-n-**di**-ny-jina*
1MIN.CRD-ERG dog-**MB** 1MIN.NOM-CM-**say**-PFV-3MIN.OBL
minyawu *yi-ngani-ny*
cat 3MIN.NOM-sit-PFV
'I thought it was a dog, but really it was a cat.' (McGregor 2022a: 481)

Whether or not the corresponding expression type in other languages also represents a separate mistaken belief construction type is uncertain; I am aware of no sources that discuss the issue. However, it does seem that the fifth observation above holds for other languages: the deictic categories of the complement clause are invariably represented in terms of an origo in the present speech situation. In other words, the complement clause seems never to be directly reported. Given that direct representation of ordinary thought complements is usually possible, this suggests that mistaken thought complements are distinct constructions.

3.2.2 A particle with an optional matrix clause of thought

Although expression by a particle is the most frequent option for non-complement expressions, only a small fraction of languages attest a particle in combination with a complement construction: Gooniyandi, Bininj Gun-wok, Kunbarlang, Burarra, Martuthunira, Yanyuwa (Pama-Nyungan), Kuuk Thaayorre and Worrorra (Worrorran) (in which language all examples given in Clendon 2014 show a matrix clause). In all but one of these languages the matrix clause employs the verb of thought ordinarily used in complement constructions. The exception is Yanyuwa, which uses a verb 'mistakenly believe' in the matrix clause together with the MB particle (see example (39) below).

Sources rarely discuss the placement of the particle. However, in most examples it occurs at the boundary between the matrix and complement clauses, as in (1) above, and (28) and (29) below. Occasional exceptions exist, such as (30), where the particle occurs initially in the matrix clause.

Djambarrpuyngu (Pama-Nyungan)

(28) *ga* *nganapurr-nydja* *nguli* ***birrka'yu****-n*
and 1PL-PROM HAB **think**-1
yanbi *nguli* *mārr* *galki,* *wānga/* *yan* *barrku* *warray*
MB HAB somewhat near place EMP far in:fact
'We thought wrongly that the place was quite close but it was far off.'
(Wilkinson 1991: 686)

Worrorra

(29) *ma-niya-ma* ***ngaali*** ***kungenga***
3M-good-3M **MB** VCOMP.1.**say**.PST
'I mistakenly thought this was a good place.' (Clendon 2014: 246)

Gooniyandi

(30) ***thaarri*** ***miga****-limi-nganggi* *manyi-yoo*
MB **think**-1SG.NOM.CL-2SG.OBL food-DAT
moow-ya *wajanginyji* *grog-joo*
look-2SG.NOM.3SG.ACC.CL.PST but:really grog-DAT
moow-ya
look-2SG.NOM.3SG.ACC.CL.PST
'I thought you were looking for food, but it was really grog you were looking for.' (McGregor 1990: 508)

It will be recalled from Section 3.1.2 that Kunbarlang and Kuuk Thaayorre employ (morphemes homophonous with) semblative particles in the marking of mistaken beliefs. These particles can occur in complement constructions, as in (31) and (32).

Kunbarlang

(31) ***yimarne*** *kadda-**ngunda**-barr* *na-bareng,* *but* *nukka* *ngorro*
MB 3PL.NF-**do**.PST-open I-dangerous but he DEM.MED.IV
karlu
NEG.PRED
'They thought he was dangerous, but he wasn't.' (Kapitonov 2021: 52)

Kuuk Thaayorre

(32) *ngay* ***kar*** ***ngeey**-m* ***kar*** *nhunt* *sixteen-nhurr-p*
1SG.NOM **MB** **hear**-PST.IMP **MB** 2SG.NOM sixteen-only-PRAG
'I thought you were only sixteen!' (Gaby 2017: 449)

Whereas use of the speech situation origo seems to be the norm for complement constructions involving clitics, things are not so clearcut in the case of complement constructions involving particles. Some languages, including Worrorra, Ngarinyin, Warray and Gooniyandi, admit at least some deictic categories in the complement clause to be represented according to the origo of the thought event. Example (1) demonstrates this for Gooniyandi, example (33) for Worrorra. In (33) the person categories of the complement clause have been shifted to the origo of the situation of the thought; the addressee of the entire utterance is represented in the second person in the matrix clause but in the third person in the complement. In this circumstance in Gooniyandi the addressee of the utterance is never represented from the person origo of the situation of thought, as shown by (22). This is as usual in reporting speech and thoughts in the language (McGregor 1990).

Worrorra

(33) *karle* *kenga* ***ngaali*** ***kunyaju**-ng-a-nu*
now 3MA.go.TNS **MB** VCOMP.1PL.EXC.**say**-PST-EP-2DAT
'We thought you'd gone (but you haven't).' More literally, 'We mistakenly thought about you$_1$ "he/she$_1$ has gone now".' (Clendon 2014: 350)

3.2.3 A suffix with an optional matrix clause of thought

Complement constructions are also found with MB suffixes (see Section 3.1.3); as expected, the suffix appears in the complement clause. Example (34) provides illustration.

Ngiyambaa

(34) *dhibiny-**dji**-dju-na* ***winanga**-nhi*
bird-**CIRC**-1NOM-3ABS **think**-PST
'I thought he was a bird.' (Donaldson 1980: 96)

Possibly also in Ngaanyatjarra, the MB suffix can occur within the complement of the 'hear, think' verb. This seems to be the case in (35), which shows both the MB suffix and the clitic. It is not entirely certain, however, whether the complement clause includes the final four words or just the first one, *palya-palya-lkanyu* 'mistakenly believed for fun'. (Unfortunately, examples of the suffix in other Western Desert varieties are unavailable.)

Ngaanyatjarra

(35) *tjilku* *pirni-lu-ya* *tjiinya* ***kuli**-ra* *palya-palya-**lkanyu***
child many-ERG-3PL.NOM you.know **think**-PRS fun-fun-**MB**
pitul-pa *ngarri-rra-**nytjakukantja**-lu* *kapi-**kukantja**-lu*
petrol-ABS lie-PRS-**MB**-ERG water-**MB**-ERG
'All the children, you know, are mistakenly thinking that petrol is lying about for fun, they mistakenly think it is (as harmless as) water.' (Glass 1983: 40)

3.2.4 An epistemic modal marker with an obligatory matrix clause of thought

Perhaps five languages of the sample show an MB construction involving an obligatory complement clause together with a morpheme that otherwise marks an epistemic modal category of uncertainty.[7] Three of these languages are geographically proximal: Ngarinyin, Warrwa (where it coexists with a construction employing an MB clitic – see Section 3.1.1) and probably Bunuba. Two more northern languages may employ this type of construction: almost certainly Dalabon (Gunwinjguan), though just a couple of examples are given in the sources; and Kuuk Thaayorre, where it is attested in a few examples, but not discussed in the grammars, and coexists with constructions involving (a homophone of) the semblative morpheme.

The construction involves two essential components: (a) a particle or enclitic that, in independent occurrence, marks an epistemic mood; and (b) a matrix clause with the generic inflecting verb 'say, think, do'. In contrast with the construction types discussed so far, the matrix clause is central to the construction, and in its absence the modal particle or enclitic expresses an epistemic meaning, as shown by (36). Examples of this construction type expressing mistaken belief are (5) (probable) and (37).

7 Recall also that in Wubuy (Nunggubuyu) there is a poorly attested expression involving two particles, one of which is either an epistemic (doubt) or a mirative (surprise) mood marker and that in Gurindji a dubitive clitic is often employed in collocation with the MB particle.

Bunuba

(36) *thurranda* *muway-**yarra*** *nyirra-yuwa* *baga-yiyirray*
two place-**PHP** that.one-LOC camp-1PL.EXC.CL.PST
'We camped there for perhaps two days (literally, two places, i.e. two camps, thus two days).' (Rumsey 2000: 104)

Warrwa

(37) *kaliya-**ngundany*** *kurd* *ngi-rnda-ny* *ø-ji-n-jina/*
finish-**PHP** die 3MIN.NOM-go-PFV 3MIN.NOM-say-PRS-3MIN.OBL
nyinka/
this
'He (the boy) still thinks it (the turtle) died.' (McGregor 2022a: 777)

McGregor (2021, 2022a) argues that this mode of expression in Warrwa represents a separate construction type, distinct from ordinary thought complements. The reasons are as outlined in Section 3.2.1 for the distinct status of the complement construction employing the MB clitic, and they are not repeated here. The same likely holds true for the other four languages; see also McGregor (2021: 11–12) on Ngarinyin. It is important to note that it is the construction as a whole that expresses the MB meaning. The mistaken belief meaning cannot be derived compositionally from the meaning of the components, the modal clitic and the thought complement.

In Dalabon an additional component is present in all putative examples of the mistaken belief construction in Ponsonnet (2013), the semblative particle *djehneng* 'as if', as in (38). Unfortunately, the source does not discuss this mode of expression, and it is unclear whether it actually codes mistaken belief. That interpretation – 'they mistakenly believe we are drinking alcohol' – is however plausible, and seems more apt than the compositional meaning expressed in Ponsonnet's free translation.

Dalabon

(38) *kardu* *ngorr* *bula-h-kurlhkurhka-wurd.* *yow,* *bah*
maybe 1PL.INC 3PL>1-R-visit.RED.PRS-DIM INTJ but
ngorr *bula-h-na-ng.* ***kardu*** *bula-h-men-yin* ***djehneng***
1PL.INC 3PL>1-R-see-P.PFV **maybe** 3PL-R-ideas-say:PRS **as.if**
kardu *ngungurru-kolh-ngu-n* *wah.*
maybe APPR.1PL.INC>3-liquid-eat-PRS water
'It seems that they [the police patrol] are coming a little towards us. Yeah, and they saw us. Perhaps they believe we might be drinking alcohol.' (Ponsonnet 2013: 114)

3.2.5 Complement clauses with a dedicated 'mistaken belief' verb

Five languages appear to show a specific verb – either a simple inflecting verb or a compound comprising an uninflecting preverbal element and an inflecting verb – glossed 'mistakenly believe' or something similar in the source: Mawng (Iwaidjan), Wagiman (isolate), Wakaya (Pama-Nyungan), Yalarnnga and Yanyuwa. The only clear examples, however, are found in Yanyuwa, which employs a verb glossed 'erroneously thinking, mistakenly thinking' (elsewhere glossed 'shake shoulders tremulously' – Bradley and Kirton 1992: 326). As in example (39), this verb seems to be invariably accompanied by the MB particle.

Yanyuwa

(39) *janda-**yudirri**-nji* ***kath**-anda* *marrwuarra* *kurdardi*
she-**mistakenly.believe**-PRS **MB**-she cousin no
wundarrba-ntha-wu *ma-kijululu*
name-PTC-DAT FD-money
'(The shop assistant) mistakenly thinks that (your female) cousin does not recognise the right change.' (Kirton and Charlie 1996: 205)

3.2.6 Complement clauses with a dedicated 'mistaken belief' nominal

In just one language, Yidiny (Pama-Nyungan), it seems that mistaken beliefs are expressed by complement clauses involving a nominal rather than a verbal predicate. The nominal is *bina*, which shows the polysemies 'ear', 'gill of fish' and 'mistaken belief'.[8] That *bina* in its 'mistaken belief' sense remains a nominal is supported by example (40), which shows its use as the predicate in a verbless clause attributing the mistaken belief to some person. This verbless clause serves as the matrix clause in the mistaken belief complement construction, in the same way as the verbal clauses discussed in previous sections.

Yidiny

(40) ***ngayu*** ***bina*** *nganyji* *gadaany* *yingu* *bama-gimbalnguri/*
I **MB** we came here man-PRV
'When we came here I thought there'd be no (Yidiny) people here.' (Dixon 1991: 118)

8 A polysemy between 'ear' and 'knowledge' is not unusual in Australian languages – see Evans and Wilkins (2000). The connection between knowledge and mistaken belief emerges via pragmatic implicature, as discussed below.

According to Dixon (1977: 137) *bina* 'mistaken belief' "is used like a particle with the meaning 'I thought something was the case but it was not'". The qualification "like a particle" is crucial: the word remains a nominal. On the other hand, as shown by (41), it may be someone other than the speaker who held the mistaken belief. (This example also employs a semblative morpheme in the mistaken belief clause; this is optional, however.)

Yidiny

(41) ***bina*** *ngayu-nguri* *mundu-uy* *nyuni-ingu*
MB I-SEM spirit-COM you-PURP
'(He) thought I was in love with you (but I'm not).' (Dixon 1977: 137)

One other language, Nyulnyul, uses a similar construction in the expression of mistaken beliefs. However this sense is a defeasible pragmatic implicature (see Section 3.3 below).

3.2.7 Concluding remarks

To wind up the discussion of mistaken belief complements, a few observations are in order. To begin with, as already mentioned, one reason for using a complement construction is to specify the holder of the mistaken belief, by coding them as the thinker in the matrix clause using cross-referencing pronominals and/or NPs identifying the thinker, as in (24), (32) and (35).

Complement constructions, in some languages at least, also permit finer temporal distinctions than are possible with a particle alone. In an expression involving just a particle, the deictic categories are always represented from the perspective of the present speech situation. In languages like Gooniyandi and Warray complement constructions can represent deictic categories in the mistaken belief clause from the origo defined by the thought itself. Thus in (42) the cloud's disappearance is set in the future with respect to the thinking event. Using the particle alone, the mistaken belief is necessarily represented temporally from the perspective of the current speech situation, in this instance past time. But that clause could potentially refer to an event in the past with respect to both the thinking event and the present. The relative times of the thought event and the event thought about can be neatly distinguished by tense choice in the complement construction.

Gooniyandi

(42) *miga-limi* *thaarri* *ngoomoorroo* *ran-**boondirni***
think-1SG.NOM.CL MB cloud leave-**FUT.3SG.NOM.CL.POT**
'I thought the cloud would go (but it hasn't yet).' (McGregor 1990: 498)

Admittedly, little is known about the representational possibilities for deictic categories in complement constructions, and few descriptions say anything. As remarked already, in mistaken belief constructions marked by a clitic in the complement clause deictic categories of the complement clause seem always to be represented from the origo of the speech situation. On the other hand, Ngarinyin and Worrorra, which use only special complement constructions, always choose the origo of the reported situation – as they also do in the reporting of speech (Rumsey 1982: 158; Spronck 2015: 1). In languages that can employ a particle together with a complement clause, things are less clear. In some languages (e.g. Gooniyandi and Warray) direct reporting is attested. There seems, however, to be a strong dispreference in Gooniyandi for representing the addressee of the utterance as a third person in the complement clause. It is not entirely clear whether the other deictic categories are also necessarily represented from the origo of the speech situation. Furthermore, it remains uncertain whether or not direct and indirect reporting of the mistaken belief actually contrast in any language.

3.3 Mistaken belief as a pragmatic implicature

A few languages express mistaken belief exclusively (in the available examples) in ordinary thought complements using the usual 'thought' verb. These include the four Pama-Nyungan languages Walmajarri, Yindjibarndi, Yingkarta and Wangkumara. Only a few examples are provided in the sources, and all of them are previous thoughts of the speaker, as in (43) and (44). In (43) the thought is specified as mistaken by the following clause. In (44) that it is mistaken is clear from the context: it is the final utterance in an exchange concerning the ownership of the meat, where the other party has successfully made a claim on it.

Yindjibarndi

(43) *ngayi* ***kurkanytya****-rna* *pawa-yi.* *ngunhaa* *wuyut*
I **think**-PST water-ACC it nothing
'I thought it [a mirage in the middle of a plain] was water. But it's nothing.' (Wordick 1982: 210)

Wangkumara

(44) *nganyja-warra* *ngathu* ***ngalka****-nga* *nhanha* *nguthi-anha*
my-POS I.ERG **think**-PST 3SG.ACC meat-ACC
'I thought this meat here was mine (though it isn't).' (McDonald and Wurm 1979: 75)

I would argue that the mistaken belief sense emerges as a pragmatic implicature (McGregor 2021). This implicature can be derived in various ways, none of which are language particular. If we compare a plain clause with that same clause as complement in a thought construction, the latter is a more marked expression, so invokes a more marked interpretation by Levinson's M principle (Levinson 2000). Thus, if we take 'it was water' and contrast it with 'I thought it was water' (as in (43)), the latter expresses a more marked meaning than the former. One possible construal is that the proposition was believed (or thought) as stated, but is in fact not true: had it been true, why qualify it as a thought?

Another take on the same example is that 'it was water' is a stronger statement on a Horn scale than 'I thought it was water'. The weaker statement implicates, by Levinson's Q principle, that the view is no longer held by the speaker. Yet another is that 'I think it is water' is stronger than 'I thought it was water', and the weaker statement implicates that the view is no longer held.

Example (44) further illustrates the operation of implicatures. Saying that the meat was the speaker's would have been inappropriate, and contrary to the previous sentence in which the speaker has acknowledged that the meat is the addressee's and that they will return it to them. By employing the more marked thought complement the speaker acknowledges that they now realise that the meat is not theirs. Framing the proposition as a thought accounts for speaker's having taken the meat in the first place. They don't just say 'it is not mine', which, though true, explains nothing.

The same implicature can arise in languages that have a means of specifying a belief as mistaken, including English and various other languages (McGregor 2022b). On the Australian continent this is attested in Martuthunira (Dench 1994: 125), which has an MB particle, and Enindhilyakwa, with an MB enclitic, as shown by (45).

Enindhilyakwa

(45) *ningi-**lyengki-yema*** *ngayuwa* *akina* *env-kv-dhaki-yadha*
1-**head-say**.PST 1 N.that N-NSR-cook-PURP
dampa
damper
'I thought it was to cook damper. [But it turns out that it is to cook stingray.]'
(van Egmond 2013: 392)

Precisely what different nuances of meaning are conveyed by the choice of the dedicated mode of expression and the implicature are uncertain.

Nyulnyul is interesting in this regard. Recall that it has an MB enclitic according to Nekes and Worms (1953, 2006). By the time of my fieldwork in the 1980s, however, there was no memory of the enclitic. Instead, a construction was consistently used with the bound nominal *-mungk* 'belief, knowledge', which takes the same person prefixes as some inalienable body part terms (see McGregor 1995). The nominal status of this word is supported by the fact that it can be used as an attribute in an attributing clause, as in (46); this usage is impossible for a particle. (The

morphological potential of the word marks it as a nominal, and distinguishes it from an inflecting verb or preverb.) Such an attributing clause may serve as the matrix clause in a complement construction, as in (47).

Nyulnyul

(46) *kinyingk* *nga-mungk* *kad* *arri* *ningarr*
DEF 1MIN-belief still not true
'That's what I thought, but it wasn't true.' (McGregor 2012: 126)

Nyulnyul

(47) *arri* *nga-mungk* *angk-in* *i-n-nyu*
not 1MIN-belief what-ERG 3NOM-CM-get
'I don't know who took it.' (McGregor 2012: 138)

In the speech of the few remaining speakers of the late twentieth century, the only mode of expressing mistaken beliefs was a complement construction with a verbless attributing matrix clause with this inflecting nominal, as in (48).[9]

Nyulnyul

(48) *warli-in* *irr-**mungk*** *juy-in* *mi-n-dam*
everyone-ERG 3AUG-**belief** 2MIN.CRD-ERG 2MIN.NOM-CM-hit
ngay-in *arriyangk*
1MIN.CRD-ERG nothing
'They all think you hit him, but I don't.' (McGregor 2012: 355)

The mistaken belief implicature arises in such examples in the same way as suggested above for verbal matrix clauses with 'think, believe'. Representing a proposition as a thought implicates a meaning that is more marked than if it had been represented by a simple clause. This may (as per the discussion above) implicate that the proposition is in fact false. This seems to have become a generalised conversational implicature of the complement construction, which by the late twentieth century became entrenched, and a new mistaken belief construction type was emergent. This was not restricted to a first-person thinker and the past tense, as shown by (48). The similar Yidiny construction (Section 3.2.6) appears to have gone a step further in entrenchment, and the nominal has taken on the semantics of mistaken belief: it now codes (in the construction) mistaken belief, which may formerly have been an implicature, as is the case in Nyulnyul.

9 It is perhaps worth remarking that the second clause in this example is not an attributing clause with *arriyangk* 'nothing' as predicate – which latter could only mean 'I am/was nothing'. Rather, the predicate *nga-mungk* '1MIN-belief' has been omitted, being given.

4 Semantics, usage and pragmatics of mistaken belief expressions

Few descriptions of Australian languages say much about the range of meanings and uses of the MB construction beyond the trivial observation that it marks someone's belief as false or mistaken (in the opinion of the present speaker at the present time). In this section I draw together what little is known or can be inferred about the meanings that may be expressed or implicated by MB constructions.

We begin in Section 4.1 by returning to an issue that has been mentioned in a couple of places in the preceding text: the identity of the holder of the mistaken belief. Next, Section 4.2 discusses one way a mistaken belief expression can be followed up by the speaker. Then in Section 4.3 we discuss discourse uses of expressions of mistaken belief. Following this, Section 4.4 makes some suggestions regarding the meanings encoded in dedicated mistaken belief constructions, and Section 4.5 raises some puzzles. The final subsection, Section 4.6, winds things up by briefly discussing mistaken belief as a modal category.

4.1 The identity of the mistaken believer

Complement constructions expressing mistaken beliefs typically provide specification of the individual who held the belief specified as mistaken: it is typically the "subject" of the matrix clause.

By contrast, the purely morphological expressions (discussed in Section 3.1) do not admit specification of a believer. We have seen that examples from various languages show a range of possibilities: the speaker of the clause, the addressee, a third person, a group including the speaker ('we'). Examination of the database reveals that the speaker is by far the most common mistaken believer across all languages of the sample, to the extent this can be inferred from the free translations provided in the sources. That is to say, mistaken belief expressions are most commonly used as corrections to the speaker's own beliefs.

Some descriptions (as mentioned already) explicitly state that the interpretation is contextually engendered; most are silent on this matter. For the bulk of Australian languages, it is not known whether there are any restrictions on who the mistaken thinker can be.

What is encoded by the non-complement expressions is no more than that in some situation someone believed a proposition which turns out to be false (in the speaker's opinion). Thus, more literal and decontextualised English translations of

the examples in Section 3.1 would involve matrix expressions like 'someone mistakenly believed', with no specification of the identity of the believer (let alone when or where they held the belief).

In fact, even this component of meaning is too specific. In some instances there is no specific individual in mind, and the mistaken belief is attributed to a generic person in a generic situation: the mistaken belief is attributed, that is, to the average person, to any (normal) member of the society. This interpretation is attested in half a dozen or so languages, including: Gooniyandi ((13) admits the interpretation 'it was/is thought I'd hit him, but really it was that other man who hit him'); Jaru (as indicated by Tsunoda's first free translation of example (49), which suggests that in the context anyone might have thought he was dead); Warlpiri (the second free translation of (16) above suggests that the woman is widely believed to be beautiful); Warrongo (Tsunoda 2011: 683); Kayardild (Evans 1995: 381); and Bininj Gun-wok (Evans 2003: 610). This meaning comes close to a semblative one (see further Section 4.4 below).

Jaru

(49) ***kulanga*** *kurnka* *nyinang-an-i*
MB dead sit-CNT-PST
'It looked as if it was dead.' Or 'I thought it was dead.' (Tsunoda 1981: 206)

4.2 Speaker's uptake

Expressions of mistaken belief are frequently – though not always – followed by a clause stating what (in the speaker's opinion) the reality actually is or was. This is illustrated in (50), where the second clause specifies what actually made the noise. Significantly, the following clause never asserts the proposition: the MB meaning is not defeasible, and is a part of the encoded meaning of the construction.

Warrwa

(50) *jurru-mirrimba* *nga-n-di-ny-jina* ***wila***
snake-MB 1MIN.NOM-CM-say-PFV-3MIN.OBL **water**
kurdiyi ***ø-na-ndi-ny***
run **3MIN.NOM-CM-get-PFV**
'I thought it was a snake; but it was just water running.' (McGregor 2022a: 653)

In instances such as this, the speaker's uptake conveys significant and unpredictable information that adds to what is known about the referent world; the same holds also for examples (13), (23) and (30). In the case of mistaken categorisations, the uptake often specifies the real category of the entity, as in examples (3), (27) and (43). In these instances the uptake serves as a replacement of the mistaken belief. In some cases, however, it conveys little in the way of news, merely restating the obvious (examples (8), (28) and (48)).

In some languages a specific morpheme may occur at the beginning of the uptake clause, as shown by (51); occasionally, as in Djambarrpuyngu, it can also occur at the end of that clause (Wilkinson 1991: 686). This morpheme expresses a contrastive meaning which is variously translated 'but', 'but really', '(but) in fact'. The classification of this morpheme differs from language to language. In some sources it is referred to as a conjunction or connective (e.g. Eather 1990; Richards and Hudson 1990: 154). In Kunbarlang the English conjunction *but* is regularly employed (Kapitonov 2021: 52; see example (31)). In other languages it seems to be a type of modal particle – e.g. Jaru (Tsunoda 1981: 206), Gooniyandi (McGregor 1990: 508), Gurindji (Meakins and McConvell 2021: 552), and Ngardi (Pama-Nyungan) (Ennever 2021: 605). In no description are either the meanings or the uses of the morpheme discussed in detail.

Jaru

(51) *karaj* *kulanga-rna* *lan-i* ***ngandu*** *kanyji*
body MB-1SG.NOM stab-PST **in.fact** leg
'I thought I was stabbing the body, but in fact I was stabbing the leg.'
(Tsunoda 1981: 206)

Such a morpheme is attested in just 21–23 languages – i.e., a little over a third of the languages attesting a mode of expressing MB. This includes languages like Walmajarri (Richards and Hudson 1990: 154) that have no dedicated mistaken belief construction. The presence of this morpheme makes explicit that the thought is mistaken. These languages are shown in Figure 4. (Of course, absence of evidence for such a morpheme is not evidence of absence.)

In Mparntwe Arrernte the focus clitic can occur on an NP representing the correct categorisation (no other marker attested), as in (3) above; and in Martuthunira it may be that the emphatic clitic is used in this way.

In Jabirrjabirr (Nyulnyulan), according to Nekes and Worms (1953: 531), alongside *-gabad* MB are also *-gabad … -gal* and *-ga … -delba*. Unfortunately, they provide almost no discussion or exemplification. The one example they do give, (52), suggests that the second component marks what really happened. (The "though" in the gloss line is theirs.)

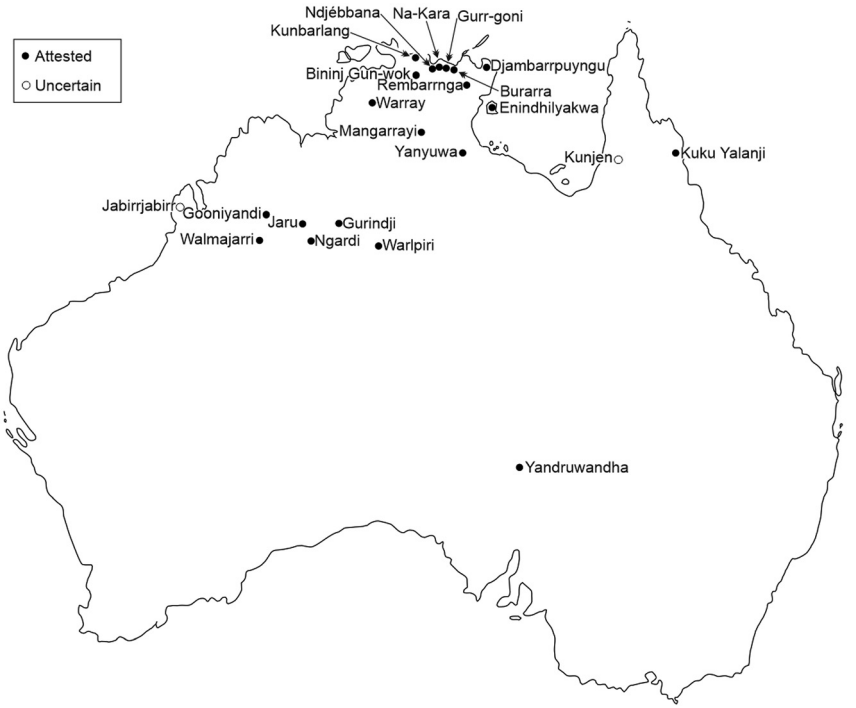


Figure 4: Languages with an attested 'but really' morpheme.

Jabirrjabirr

(52)

ŋai	*djān-djer*	***gabad***	*ŋan-dj,*
ngayi	*jaan-jirr*	***kabad***	*nga-n-j*
1MIN.CRD	1MIN.OBL-EMP	**MB**	1MIN.NOM-CM-say
dje-djer	***gal***	*djoe*	
jii-jirr	***kal***	*juy*	
2MIN.OBL-EMP	"though"	2MIN.CRD	

'I thought it was mine although it is yours.' (Nekes and Worms 1953: 531)

4.3 Discourse uses of MB expressions

Little is known about the motivation for using MB expressions in discourse, and the meanings that might be intended. Perhaps the most common motivation seems to be

to explain behaviour that is unexpected given the interactional context or the referent world and the participants in it.

A number of examples cited above illustrate this. In (25), the misclassification of the birds as water hens is used to explain why the speakers killed the chickens. Examples (26) and (37) come from a description of a drawing in the picture book *A boy, a dog, a frog and a friend* (Mayer and Mayer 1971) that depicts the boy continuing to dig a hole in which to bury the turtle, which is now clearly alive. The unexpected behaviour of the boy is accounted for by attributing to him the belief that the turtle is dead. And in (38) the behaviour of the police, surprising given that the speaker and cohorts are merely talking together, is accounted for in terms of the belief that they were drinking.

It seems likely that in examples such as (25) the explanation of unexpected behaviour might be in turn employed to excuse that behaviour: one expects that (25) could be employed by the speaker as an excuse for their killing the chickens, not just to explain it. Similarly, the Wangkumara example (44) was clearly used as an excuse in the exchange from which it comes (though in this instance the mistaken belief sense emerges via implicature). Jabirrjabirr example (52) quite likely could have been employed in the same way as an excuse.

By contrast, (23) could potentially be employed as an accusation: that the addressee has failed to perform some action expected of them.

In at least one language, Mparntwe Arrernte, the MB enclitic is sometimes employed in criticisms (Wilkins 1989: 409). This is illustrated by example (53), where the speaker is criticising the hearer for not giving them money.

Mparntwe Arrernte

(53) *tyew-atye-**kathene*** *ayenge* ***itirre**-ke*
friend-1SG.POS-**MB** 1SG.NOM **think**-PC
'I thought you were my friend.' (But you can't be since a friend would give me money.) (Wilkins 1989: 409)

Wilkins (1989: 426) suggests that the sense of criticism arises as a pragmatic implicature. I agree, though I have a slightly different account of how this meaning is implicated (elaborating somewhat on McGregor 2021: 13). To begin with, the speaker claims that the proposition P ('you are my friend') is mistaken in the face of the fact that P is clearly true, and known to be true by both interactants. Their general knowledge thus conflicts with what is stated, namely that P is not true. The speaker flouts the Gricean maxim of quality. This is for a reason. Invoking the counterfactual proposition ¬P the speaker effectively instructs the hearer to infer how it might be that they can claim P to be false. The answer is evident from the interactional context, the situation that the addressee has refused the speaker money, which by ordinary understanding would be the behaviour of people who are not friends. It follows that

you are remiss in not giving me money, and the criticism emerges via pragmatic implicature.

This is clearly an effective strategy for indirect criticism, and one would expect it to be widely employed. However, Mparntwe Arrernte is the only Australian language in which this pragmatic implicature is actually attested.

4.4 Mistaken belief and the notion of similarity

We have seen that in some languages there is an affinity between the senses of mistaken belief and similarity. As discussed in Section 4.1, the mistaken belief is sometimes not attributed to anyone in particular, in which case the sense of likeness may emerge, as shown by (54) and (55). In fact, most examples of the MB particle in non-initial position in Warlpiri invoke this sense, and (as is clearly the case in (55)) the preceding material is not included in what is mistakenly believed (see Laughren 1982: 149).

Bininj Gun-wok

(54) ***yimarnek*** *man-bedje* *la* *wayarra* *ka-bardrohrokme*
MB III-spear.grass CON spirit 3-stand.like.spear.grass.NP
'It looked like spear grass, but it really was a spirit standing up like spear grass.' (Evans 2003: 610)

Warlpiri

(55) *kala* *yakarra-pardija* *yali* ***kulanganta*** *ngurrju* *kala* *nyurnu*
AUX awake-rise.P that **MB** good but dead
yalumpu-ji
that-DEL
'That one got up as though he was fine (=alive), but he was dead.' (Laughren 1982: 149)

In these languages the semblative sense can be accounted for as a contextualisation of the coded meaning 'it was mistakenly believed': that the proposition is something that could have been believed by someone in the relevant circumstances suggests that what actually happened was like something else (namely what it could have been misconstrued as). The semblative sense is thus not emically distinct from the mistaken belief sense. Glosses like 'seemingly' – as per for example Waters (1989: 157) for Djinang – are thus appropriate, provided that they are understood to simultaneously code 'but in fact not'. It is surely significant that in all instances where this meaning comes to the fore, just a particle or clitic is found, never a complement construction. It can nevertheless be present, albeit backgrounded, in the latter

constructions: for example in (23), that something happened that looked to the addressee like the speaker went for water.

I am not suggesting that in all languages showing similarities in the expression of the mistaken belief and semblative senses the relevant clitic or particle codes the MB meaning, with similarity a contextual sense. Recall that in Sections 3.1.1 and 3.1.2 we mentioned languages in which clitics or particles of mistaken belief were (homophonous with) semblative morphemes. I suspect that these languages have a single morpheme that codes a meaning we might gloss 'like'. The mistaken belief sense can be accounted for in two ways. First, as Kapitonov and Gentens (2018) observe, it can plausibly arise via a Gricean implicature of the coded semblative meaning: what is similar is not identical. Second, as per McGregor (2021: 15), the mistaken belief sense arises when the morpheme holds a clause in its scope, modifying it propositionally. Correspondingly, the semblative sense emerges when the notion of likeness applies not to a proposition but instead to some thing or event in the discourse world. It is not propositions but items of the world of experience that show resemblances. Put in functional grammar terms, for mistaken beliefs the modification is at the interpersonal level; for semblatives it is at the experiential or representational level (Halliday 1985; Hengeveld and Mackenzie 2008).

These remarks show that neither meaning, mistaken belief nor semblative, can be presumed synchronically more fundamental or diachronically prior. Both are equally plausible candidates, and require an evidential basis.

4.5 Puzzles

Tsunoda (2011: 677) makes the astute observation that in Warrongo there is no instance in which the mistaken belief is a negated proposition: there are no instances like 'I thought he was not going, but in fact he did go'. And indeed, I have come across very few instances of this type in the descriptions. Dixon (1991: 117–119) cites a few examples in Yidiny, though they all involve the privative marker on a nominal, rather than clausal negation. A probable instance is example (56) from Yir Yoront, though the interpretation is unclear due to lack of glossing in the source.

Yir Yoront (Pama-Nyungan)

(56) ***kar**-kar* *ng'n* *invnh*
MB-not don't:know be:sitting
'[Excuse me for intruding,] but I thought you weren't here.' (Alpher 1991)

I suspect that there is a reason for the paucity of examples like (56). As is well known, negative clauses are distributionally marked and invoke presupposition of the corresponding positive (e.g. Givón 1984: 324; McGregor 1997: 226–227; Quirk et al. 1985: 808–809). It seems reasonable to hypothesise that the same holds for one's inner speech: one would think a negative proposition only when there was a presumption that the positive proposition was true. That is to say, one would think 'she is not here' in the context that they expected her to be here; otherwise, one would expect the person to entertain a positive proposition such as 'she has gone' or 'she is over there'. But then the presupposition is precisely what would be asserted if the thought was claimed to be false in a mistaken belief construction: saying 'it was mistakenly believed that she is not here' asserts that she is here, which is precisely the presupposition of the thinker. The normal expectation in these circumstances would be that one would simply state the proposition: 'she is here (just as expected)'. It seems pointless to muddy matters by invoking the thought 'she is not here', which turns out to be false.

The rub is that to do so, to use the mistaken belief construction is possible, albeit a more marked expression than the simple statement of the positive clause. This must be for a reason. We can account for example (56) as follows. By invoking the mistaken thought 'you are not here' the speaker accounts for their own presence on the scene, for why they are now in the presence of the addressee. The speaker thereby excuses themself for imposing on the addressee.

One additional example is significant, (57), the interpretation of which is far from clear. Evans (2003: 611) takes it to be a counterexample to the MB sense of the particle *yimankek*, and avers that 'they reckoned' (given in the translation by a native speaker) is an appropriate free translation. It is common knowledge among the Bininj Gun-wok that the Rainbow Serpent will not hear mosquitoes being slapped, provided it is done loudly. How then can (57) present this as a mistaken belief? I have suggested (McGregor 2021: 13) that an explanation may be possible in terms of a weakening of the second component of meaning of the MB construction (see Section 4.1. above). Specifically, the belief is false not according to the speaker, but rather in relation to general knowledge of the real world – according to which slapping mosquitoes hard would result in being heard.

Bininj Gun-wok

(57) *birndu* *kum-mirnde-wam,* *wanjh* *bene-h-bu-ni*
mosquito 3hither.PST-many-go.P.PFV then 3UAUG>3PST-IMM-hit-IMP
wernkih *ka-mak,* ***yimankek*** *Ngalyod* *minj* *benbene-bekka-yinj* *bu*
hard 3-good **MB** Rainbow not 3>3DU.PST-hear-IRR SUB

wernkih *bene-h-bu-ni.* *anjh* *yerre*
hard 3UAUG>3PST-IMM-hit-IMP then after
bene-h-yirrm-i *nawu* *birndu* *wanjh*
3UAUG>3PST-IMM-brush-IMP MA.REL mosquito then
benmene-bekka-ng *wanjh* *benmene-kuk-nguneng*
3>3DU.PST-hear-P.PFV then 3>3DU.PST-body-eat.P.PFV
'Mosquitoes started to swarm around them, and they slapped at them hard, but that was OK, they reckoned Ngalyod couldn't hear them slapping hard. But when they started brushing them off gently, he heard them and ate the two of them.' (Evans 2003: 611–612)

Another explanation is possible. If a Bininj Gun-wok speaker had wanted to express a mistaken belief in this context, one would have expected them to have said 'They mistakenly believed Ngalyod would hear them if they slapped the mosquitoes hard'. But this could hardly be true: why would they have hit the mosquitoes hard if they believed the rainbow serpent would hear them – and consequently eat them? Perhaps the speaker is negating the entire proposition 'They mistakenly believed that the rainbow serpent would hear them if they slapped the mosquitoes hard'. This would mean that the negative particle holds the entire sentence in its scope, not just the proposition mistakenly believed. The problem is that this seems to conflict with the scopal relations as suggested by the ordering of the two particles. This is not necessarily an insuperable problem: ordering of elements does not always reflect scoping relations – see also example (56), where the negator appears to be in the matrix clause rather than the complement clause, as predicted by the scoping relation.

It is impossible to decide among the above analyses. My purpose is simply to show that the mistaken belief interpretation need not be ditched out of hand.

4.6 Concluding remarks: mistaken belief as a modal category

We have seen that some Indigenous languages of Australia have a morpheme or construction that codes mistaken belief: this meaning is invariably present and is indefeasible. In such languages, mistaken belief is an epistemic modal category that invokes a dual perspective on a proposition. On the one hand, they indicate that the proposition was believed by someone at some point in time, or could have been believed by someone in the circumstances. On the other hand, they indicate that the proposition is in fact false from the perspective of the present speaker in the here and now. Examples like (53) do not contradict the second component of meaning. Quite the contrary: the implicature relies on the fact that the encoded meaning is contrary

to the facts as understood by the speaker. However, as discussed in Section 4.5., it may be that this second requirement also needs to be relaxed somewhat: the proposition might instead be evaluated from the perspective of general knowledge of the world.

As an epistemic modal category mistaken beliefs concern propositions, things that can be true or false, not occurrences of events. Thus I have proposed (Section 3) that in circumstances in which just the category of some entity is at issue, the marker holds a full categorising clause in its scope, not just a nominal. An important fact about mistaken belief constructions is that they represent more than mere thoughts. The thinker does more than simply entertain the thought: they also believe it. One could easily imagine they were living on the moon without mistakenly believing that they were. Hence in contrast with ordinary thought complements mistaken beliefs never comprise just an interjection, and indeed seem never to contain one. This suggests the distinctiveness of expressions coding mistaken belief as constructions separate from reported thoughts, extending on the remarks in Sections 3.2.1. and 3.2.4. Fuller arguments are called for.

The fact that mistaken belief constructions marked by just a morpheme seem generally to be non-specific as to the person holding the mistaken belief underlines their comparability with other modal categories. For instance, both epistemic and deontic senses of *must* in English admit a similar range of possibilities for the source of the epistemic or obligative meaning: the speaker, some key third person, a general member of the society, even the addressee themself.

It has emerged in the discussion above that there are sometimes similarities in the marking of mistaken belief and other meanings such as similarity (discussed in Section 4.4.) and modal categories such as counterfactuals and irrealis. It is beyond the scope of this paper to explore the similarities and differences among the various modal categories. Suffice it to say that while mistaken belief resembles these categories in terms of the fact that it marks the proposition as false, it conveys an additional meaning not conveyed by either counterfactual or irrealis, namely that someone (might have) believed the proposition.

5 Conclusion

This paper has surveyed the range of attested modes of expressing mistaken beliefs in a largish set of Australian languages. The most serious problem facing any typological investigation of this domain is lack of data. Few grammars provide any discussion of the modes of expression, and those that do not give very comprehensive treatments: issues such as word order, presupposition, scope and focus are rarely discussed at all, let alone in depth. We need much broader and deeper treatments of the phenomenon in Australian languages.

I have not discussed the grammatical structures involved in mistaken belief constructions, though I have elsewhere made some preliminary suggestions (McGregor 2021: 15). In short, what I have proposed is that in the constructions involving just clitics and particles, this morpheme holds the remainder of the clause within its scope, modalising the proposition it expresses. This also holds true when a 'like' morpheme is employed (such morphemes enter into different types of grammatical relations when used as semblatives). In complement constructions the matrix clause may frame the complement clause, specifying its status as a demonstration or depiction of the proposition as it might be expressed in words, as per Clark and Gerrig (1990) and Clark (2016). Sometimes a particle or clitic simultaneously scopes over the proposition, and sometimes the matrix clause may scope over the complement clause. The crucial observation here is that in all cases a comparable grammatical relation is involved, a relation of the conjugational type (McGregor 1997, 2021).

Among the many issues demanding further investigation, a few are worth highlighting.

First, the semantics, contextual meanings and pragmatics of mistaken belief expressions in all languages are poorly understood. We have little knowledge of what meanings are contextually engendered or implicated rather than encoded, and very little concerning presuppositions. The discourse uses of mistaken belief expressions are also in need of further study.

Second, the diachrony of mistaken belief expressions has barely been dealt with either in the literature or in this paper. However, some of the observations above suggest possible grammaticalisation scenarios. For instance, we have already commented on the possibility that the Yidiny mode of expressing mistaken beliefs derives from something like the Nyulnyul one via entrenchment. It is not implausible that a mode of expression such as the Yidiny one could give rise to a mistaken belief particle. This suggests that we might find cognates of MB particles in nominals meaning 'knowledge' – possibly also verbs. Another possibility is that a morpheme 'like' could, by virtue of being employed in a scoping relation with a clause, become an MB enclitic if its other uses became obsolete.

Third, more generally, the entire domain of representation of thought and cognition in Australian languages demands more thorough investigation. McGregor (2021) is a beginning, and I have started an investigation of the typology of expressions of knowledge and ignorance in Australian languages. Paucity of data is again a problem, though of slightly lesser magnitude than in the case of mistaken beliefs.

Fourth, it is worth extending the typological investigation of mistaken belief expressions beyond Australian languages. Together with Caroline Gentens and Stef Spronck, we have recently begun planning such a project, which is again hampered by descriptive inadequacies. At present it can only be remarked that there is evidence of particles/clitics of mistaken belief as well as complement constructions in

some South American languages (Spronck and Vuillermet 2019), and that complement constructions predominate in languages of Eurasia (McGregor 2022b).

Abbreviations and conventions

The following abbreviations are employed in the representation of examples:

ABL	ablative
ABS	absolutive
ACC	accusative
ALL	allative
APPR	apprehensive
AUG	augmented
AUX	auxiliary
CIRC	circumstantive
CL	classifier
CM	conjugation marker
CNT	continuous aspect
COM	comitative
CON	conjunction
CRD	cardinal
DAT	dative
DEF	definite
DEL	delimiting
DEM	demonstrative
DIM	diminutive
DU	dual
EXC	exclusive
EMP	emphatic
EP	epenthetic
ERG	ergative
FD	food gender
FOC	focal
FUT	future
GEN	genitive
HAB	habitual
IMM	immediate
IMP	imperfective
INC	inclusive
INF	infinitive
INTJ	interjection
IRR	irrealis
LOC	locative
M	M gender
MA	masculine gender
MB	mistaken belief

MED	medial
MIN	minimal
N	neuter gender
NEG	negative
NF	non-future
NOM	nominative
NP	non-past
NSR	nominaliser
OBL	oblique
PC	past completive
PFV	perfective
PHP	perhaps
PL	plural
POS	possessive
POT	potential
P.PFV	past perfective
P.PTC	past participle
PR	prereferential noun
PRAG	pragmatic enclitic
PRED	predicative
PROM	prominence
PRS	present
PRV	privative
PST	past
PTC	participle
PURP	purposive
R	realis
RED	reduplicated
SEM	semblative
SG	singular
SUB	subordinator
TNS	tense
UAUG	unit augmented
VCOMP	sentential complement index
XTD	extended action

The first three integers indicate person categories, roman numerals I-IV are used to indicate gender classes in some languages, and > means 'acting on'. Inflecting verbs in languages with small classes of verbs are cited in capitals, following a convention I have employed elsewhere. It is often impossible to be certain – given the information in the sources – of the division of words into morphemes and/or the appropriate glosses for the morphemes. I have usually avoided analysing examples further than in the sources. Thus in some cases morphologically complex inflecting verbs are not divided into morphemes, but treated as portmanteaus and given a sequence of glosses separated by periods. Periods are thus used in gloss lines to

separate glosses regardless of whether or not they belong to portmanteau forms. For clarity, parts of example sentences relevant to the discussion are bolded.

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